Deep Learning for Computer Vision

Regularization in Neural Networks

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Review

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- Let us assume a linear deep neural network (only linear activation functions, $f(x) = x$). Would using GD and MSE still be a non-convex problem?

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- Let us assume a linear deep neural network (only linear activation functions, $f(x) = x$). Would using GD and MSE still be a non-convex problem? Yes! Weight symmetry is a reason

What is Regularization? An Intuitive View

What's my rule?

- 1 2 3 $\implies$ satisfies rule
- 4 5 6 $\implies$ satisfies rule
- 7 8 9 $\implies$ satisfies rule
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Plausible rules

- 3 consecutive single digits
- 3 consecutive integers
- 3 numbers in ascending order
- 3 numbers whose sum is less than 25
- 3 numbers < 10
- 1, 4, or 7 in first column
- "yes" to first 3 sequences, "no" to all others

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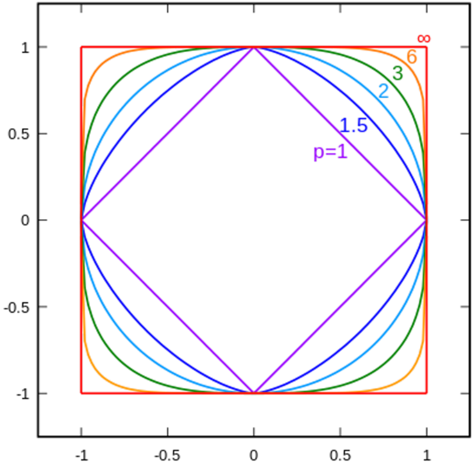
Regularization corresponds to methods used in machine learning to improve **generalization performance** (avoid overfitting to training data)

Review: L_p Norms



- All p-norms penalize larger weights
- $p < 2$ tends to create sparse (i.e. lots of 0 weights)
- $p > 2$ tends to like similar weights

Credit: David Kauchak, Pomona Univ

L2 Regularization

- Penalty on L2-norm of parameters commonly known as L_2 **weight decay**, or simply **weight decay**
- Loss function defined by:

$$\widetilde{\mathcal{L}}(w) = \mathcal{L}(w) + \frac{\alpha}{2} \|w\|^2$$

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- Update rule:

$$w_{t+1} = w_t - \eta \nabla \mathcal{L}(w_t) - \eta \alpha w_t$$

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$$\mathcal{L}(w^* + u) = \mathcal{L}(w^*) + u^T \nabla\mathcal{L}(w^*) + \frac{1}{2} u^T H u$$

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- Gradient of total objective function (in presence of L2 regularization):

$$\nabla\widetilde{\mathcal{L}}(w) = \nabla\mathcal{L}(w) + \alpha w = H(w - w^*) + \alpha w$$

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&= (Q\Lambda Q^T + \alpha Q\mathbb{I}Q^T)^{-1}Q\Lambda Q^T w^* \\
&= [Q(\Lambda + \alpha\mathbb{I})Q^T]^{-1}Q\Lambda Q^T w^* \\
&= Q^{T^{-1}}(\Lambda + \alpha\mathbb{I})^{-1}Q^{-1}Q\Lambda Q^T w^* \\
&= Q(\Lambda + \alpha\mathbb{I})^{-1}\Lambda Q^T w^* (\because Q^{T^{-1}} = Q)
\end{aligned}$$

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- Summary: The weight vector (w^*) is getting rotated to $\widetilde{w}$ on using L2 regularization. All of its elements are shrinking but some are shrinking more than the others. This ensures that only important features are given high weights.

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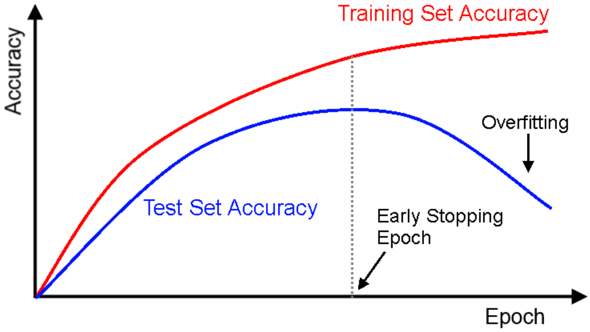

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- Helps increase data; is useful when training data provided is less (DNNs need large amounts of training data to work!)
- Also acts as regularizer (by avoiding overfitting to provided data)

Dataset Augmentation: Example[1]

[1]Wu, Ren, et al. "Deep image: Scaling up image recognition." arXiv 2015

Dataset Augmentation: Example[1]

Left rotation, crop | Right rotation, crop | Pincushion distortion | Barrel distortion

Horizontal stretch | More Horizontal stretch | Vertical stretch | More vertical stretch

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Data Augmentation: Newer Methods

Mixup

- Create virtual training examples as below ($\lambda \in [0, 1]$):

$$\tilde{x} = \lambda x_i + (1 - \lambda)x_j$$

$$\tilde{y} = \lambda y_i + (1 - \lambda)y_j$$

  where x_i,x_j are input vectors, and y_i,y_j are one-hot label encodings
- *Variants:* Manifold Mixup, AugMix

CutOut

- Randomly mask out square regions of input during training

- *Variants:* CutMix

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- **Label Noise**
- **Gradient Noise**

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Regularization through Label Noise[3]

- Disturb each training sample with probability α.
- For each disturbed sample, label randomly drawn from uniform distribution over $\{1, 2, \cdots, C\}$, regardless of true label

Algorithm 1 DisturbLabel

1: **Input:** $\mathcal{D} = \{(\mathbf{x}_n, \mathbf{y}_n)\}_{n=1}^{N}$, noise rate α.
2: **Initialization:** a network model $\mathbb{M}$: $\mathbf{f}(\mathbf{x}; \boldsymbol{\theta}_0) \in \mathbb{R}^C$;
3: **for** each mini-batch $\mathcal{D}_t = \{(\mathbf{x}_m, \mathbf{y}_m)\}_{m=1}^{M}$ **do**
4:     **for** each sample $(\mathbf{x}_m, \mathbf{y}_m)$ **do**
5:         Generate a disturbed label $\widetilde{\mathbf{y}}_m$ with Eqn (2);
6:     **end for**
7:     Update the parameters $\boldsymbol{\theta}_t$ with Eqn (1);
8: **end for**
9: **Output:** the trained model $\mathbb{M}'$: $\mathbf{f}(\mathbf{x}; \boldsymbol{\theta}_T) \in \mathbb{R}^C$.

$$\left\{ \begin{array}{rcl} \widetilde{c} & \sim & \mathcal{P}(\alpha), \\ \widetilde{y}_{\widetilde{c}} & = & 1, \\ \widetilde{y}_i & = & 0, \qquad \forall i \neq \widetilde{c}. \end{array} \right. \qquad (2)$$

[3] Xie, Lingxi, et al. "DisturbLabel: Regularizing CNN on the Loss Layer." CVPR 2016.

Regularization through Gradient Noise[4]

- **Idea:** Add noise to gradient

$$g_t \leftarrow g_t + N(0, \sigma_t^2)$$

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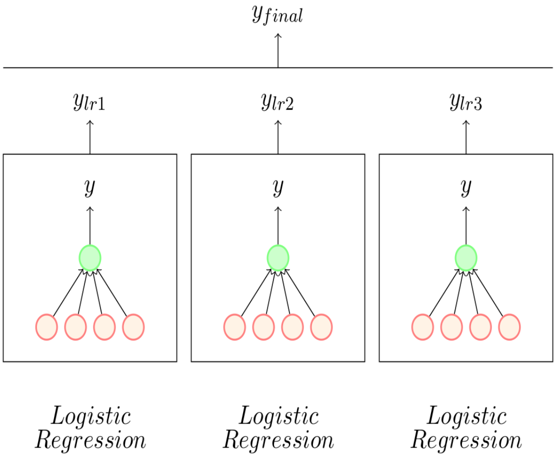
Ensemble Methods

- Train several different models separately, then have all models vote on the output for test examples
- An example of a general strategy in machine learning called **model averaging**
- Models can correspond to different classifiers, or could be different instances of same classifier trained with:
  - different hyperparameters
  - different features
  - different samples of the training data

Credit: Ali Ghodsi, Univ of Waterloo; Mitesh Khapra, IIT Madras

Ensemble Methods

- **Bagging** (short for bootstrap aggregating): reduces generalization error by forming an ensemble using different instances of same classifier
- For e.g., consider a set of k logistic regression models
- Given a dataset, construct multiple training sets by sampling with replacement $(T_1, T_2, ..., T_k)$
- Train i^{th} instance of classifier using training set T_i



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- Expected squared error of ensemble predictor is:

$$\begin{aligned}
MSE &= \mathbb{E}\Big[\big(\frac{1}{k}\sum_i \varepsilon_i\big)^2\Big] \\
&= \frac{1}{k^2}\mathbb{E}\Big[\sum_i \sum_{j=i} \varepsilon_i \varepsilon_j + \sum_i \sum_{j \neq i} \varepsilon_i \varepsilon_j\Big] \\
&= \frac{1}{k^2}\mathbb{E}\Big[\sum_i \varepsilon_i^2 + \sum_i \sum_{j \neq i} \varepsilon_i \varepsilon_j\Big] \\
&= \frac{1}{k^2}(kV + k(k-1)C) \\
&= \frac{1}{k}V + \frac{k-1}{k}C
\end{aligned}$$

Credit: Ali Ghodsi, Univ of Waterloo; Mitesh Khapra, IIT Madras

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- Expected squared error of ensemble predictor:

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- If errors of model are independent or uncorrelated, then $C = 0$ and MSE of ensemble reduces to $\frac{1}{k}V$
- On average, **ensemble will perform at least as well as its individual members**

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Dropout

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  - As network is trained iteratively, powerful connections are learned more while weaker ones are ignored
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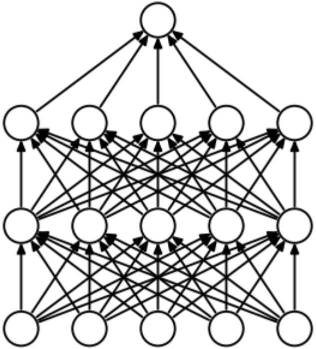
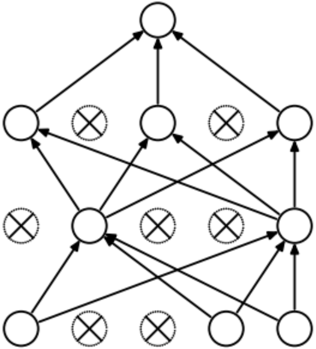
Dropout



(a) Standard Neural Net



(b) After applying dropout. [5]

[5]Srivastava, Nitish, et al. "Dropout: a simple way to prevent neural networks from overfitting", JMLR 2014

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NGM is equivalent to output of overall network with weights divided by two

Homework

Readings

- Deep Learning book, Chapter 7, Sections 7.1-7.5, 7.8, 7.12
- Tutorial on Dropout for more information (Optional)

Exercise

- Show that adding Gaussian noise (with zero mean) to input is equivalent to L_2 weight decay when loss function is MSE.

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